Read the Boss

Stamina-constrained tactical combat | Qipan Xu

1. Project Overview

- **Idea:** A turn-based 1v1 duel tests whether an RL fighter can manage stamina and transfer its decisions to unfamiliar boss behaviors.
- **Implemented:** A custom Gymnasium environment, seven scripted boss modes, two reward variants, legal-action masks, a visual renderer, human terminal play, and three heuristic policies. Automated tests and reproducible execution evidence are included.
- **Remaining:** Q-learning/SARSA training, baseline and difficulty calibration, equal-budget robustness experiments, and final reward analysis. No trained RL policy is claimed here.

2. Environment Design

Environment. One fighter (6 HP, 4 maximum stamina) faces an 8-HP boss at close, medium, or far distance. Episodes end at victory, death, or the intrinsic 40-turn deadline. Balanced, aggressive, and distance-aware controllers plus their per-turn mixture are implemented; three additional controllers are reserved for later policy evaluation.

State / observation. The integer vector is **[player HP, boss HP, stamina, distance, intent, turns remaining]**. Reset returns **[6, 8, 4, 1, 3, 40]**; distance codes 0/1/2 mean close/medium/far, and intent codes 0/1/2/3 mean slash/sweep/lunge/recovery. The observation includes the visible next attack and remaining time; no hidden persistent boss identity is used.

Actions. Approach moves closer for free; retreat moves away for 1 stamina; strike costs 2 and deals 2 damage when close; guard costs 1 and blocks up to 2 damage; rest restores 2 stamina, capped at 4. All supplied policies and the human interface obey action masks.



Figure 1. Actual RGB rendering immediately after `reset(seed=7)`. Strike is dimmed because the fighter starts at medium distance. The schematic uses discrete distance bands.

2. Environment Design (continued)

Reward. Every step gives $-0.1 - 0.02c$, where c is stamina spent. Victory adds $100 + 10h/6$ using remaining player health; death or timeout adds -100. Waiting/resting earns no bonus. The optional terminal/time-only mode removes the health and stamina terms.

Transition. Apply the player action and stamina cost, then resolve the previously announced boss intent at the new distance. A lethal player strike cancels retaliation. Slash deals 2 damage at close range; sweep deals 3 at close/medium; lunge advances one band then deals 1 if close; recovery deals none. Decrease the clock, check victory/death before timeout, and sample the next visible intent only if combat continues.

read_the_boss/env.py | step() after input validation

```
141  action = Action(int(action))
142  state = list(self.state)
143  events: list[str] = []
144  invalid = not bool(self.action_masks()[action])
145  spent, dealt, guard = self._apply_player_action(state, action, invalid, events)
146  taken = self._resolve_boss(state, guard, events) if state[1] > 0 else 0
147  if state[1] == 0:
148      events.append("Lethal strike: boss retaliation cancelled.")
149  state[5] -= 1
150  outcome = self._outcome(state)
151  self._terminated = outcome != "ongoing"
152  if not self._terminated:
153      state[4] = choose_next_intent(self.opponent, state, self.np_random)
154      events.append(f"Next intent: {Intent(state[4]).name.lower()}.")
155  else:
156      events.append(f"Episode ended: {outcome}.")
157  self._state = tuple(state)
158  terms = self._reward_terms(outcome, spent)
159  reward = float(sum(terms.values()))
160  info = self._info(outcome, invalid, events, spent, dealt, taken, terms)
161  self._last_info = deepcopy(info)
162  return self._observation(), reward, self._terminated, False, info
```

Figure 2. Verbatim source excerpt from `read_the_boss/env.py`, lines **141–162**. Full validation and helper methods are in the source ZIP.

The excerpt shows the executed transition order: mask checking, player action, conditional boss retaliation, clock/outcome update, and stochastic next intent. Reward terms are summed explicitly before returning Gymnasium's observation/reward/termination/info tuple.

3. Environment Testing

- **33/33 tests passed:** Gymnasium API/render checks, reset/reseeding, observation bounds, masks, stamina, attack ranges, guard effects, rewards, and victory/death/timeout ordering.
- **Reset and action checks:** reset restores the six initial values and masks; identical seeds/actions reproduce trajectories. Controlled cases verify lethal-hit priority, last-turn timeouts, zero stamina, and invalid-action handling.
- **Constraints and uncertainty:** 280 complete random-action episodes check invariants; 60,000 sampled intents check controller probabilities. Mixed rules are resampled each turn, so stochastic behavior does not introduce an unobserved persistent opponent type.

3. Environment Testing (continued)

Measured execution output | complete logs in evidence/

```
$ python -m unittest discover -s tests -v
PASS 33/33 tests; failures=0; errors=0
Gymnasium checker + seeded combat/range/resource tests

$ python -m read_the_boss demo --seed 7 --policy reactive
ORDER: [player HP, boss HP, stamina, distance, intent, turns]
RESET [6, 8, 4, 1, 3, 40]
t=01 APPROACH [6, 8, 4, 0, 1, 39]  r=-0.100
t=02 GUARD    [5, 8, 3, 0, 3, 38]  r=-0.120
t=03 STRIKE   [5, 6, 1, 0, 2, 37]  r=-0.140
... full 10-turn trace: evidence/demo/trace.txt
END VICTORY | HP=1 | return=+100.467

SMOKE 1,200 heuristic episodes | 10,834 steps | 0 masked actions submitted
No trained RL policy; no held-out policy benchmark.
```

Figure 3. Summary rendered from actual test, demo, and smoke outputs. Full records: `evidence/tests.txt`, `evidence/demo/trace.json`, and `evidence/smoke_results.json`. The abbreviated trace omits intermediate turns, not failures.

Observed action effects. In the seed-7 demonstration, approach changes medium to close and reduces the clock to 39 (reward -0.10). The following guarded sweep leaves player HP at 5 and stamina at 3 (reward -0.12); the guard blocks 2 of its 3 damage. The fixed reactive heuristic wins this episode in **10 turns**, with **1 HP** and return **100.467**.

Broader execution check. Three untrained policies were run for 100 episodes per training mode, using shared seeds 1000–1099. All 1,200 episodes completed within 40 turns; the run used 10,834 transitions and submitted 0 masked actions. Measured win rates were:

Boss controller	Random	Aggressive	Reactive
Balanced	0%	22%	22%
Aggressive	0%	8%	13%
Distance-aware	0%	20%	56%
Mixed	0%	20%	36%

These small smoke checks demonstrate execution, not learned-policy quality or the benefit of mixed training. Held-out mechanics are unit-tested, but held-out policy benchmarking remains reserved. Optional animated replay: `evidence/demo/demo.gif` (actual recorded frames).

4. Next Steps

- **Implement:** masked Q-learning and SARSA, equal-step-budget single/mixed training, multiple training seeds, and the reward ablation. Freeze the design and validation choices before benchmarking on reserved enemy behaviors.
- **Current problem:** combat is demanding: reactive-heuristic win rates range from 13% to 56% in these checks. Calibrate the baseline and difficulty on development/validation episodes, document any rule changes, and establish learning progress before final evaluation.